MedTrack

Project Pitch | DCDS Lab ESE Miniproject | Team of 4

The Problem

Missing a medicine dose, forgetting to drink enough water, or not knowing that two prescribed drugs interact badly are some of the most common — and most preventable — causes of poor health outcomes. This hits elderly patients and people managing multiple prescriptions the hardest, and existing solutions are either too generic (a plain alarm app with no medical context) or require a hospital visit even for routine issues that don't need one.

The Idea

MedTrack is a Python-based application that turns a doctor's prescription into a structured daily routine and gives patients, nurses, and doctors a shared system to manage it together, instead of leaving adherence entirely up to the patient's memory. It has three connected roles:

- **Patient** — gets reminders for medicine and water intake, logs each dose as taken or missed, keeps a symptom journal, and can raise an SOS if something feels wrong or they've run out of medicine.
- **Nurse** — permanently assigned to a set of patients, acting as the first responder to any SOS: handling routine issues directly (a delivery, a scheduled shot) and escalating only what actually needs a doctor's judgment. This is what makes the system practical — doctors don't have time to review every complaint themselves.
- **Doctor** — creates and manages prescriptions, gets warned automatically if a new prescription conflicts with one the patient is already on, assigns a nurse to each patient, and only sees the SOS cases that were escalated to them.

Why It Fits This Lab

The project is built entirely on synthetic data, so it stays within the scope of a lab miniproject while still exercising every part of the evaluation: a working Python application, a relational database (MySQL) for structured records like patients, prescriptions and dose logs, a NoSQL database (MongoDB) for flexible data like symptom notes and hydration logs, advanced querying (joins, aggregations, a stored procedure for auto-marking missed doses), and real validation/error handling around things like dosage input and drug conflicts.

The Dataset

Since this is an academic project using no real patient data, the dataset is built in two parts:

- **Synthetic operational data** — generated with Python's Faker library to populate patients, doctors, nurses, prescriptions, dose logs, and SOS/task records (around 25 patients, 4 doctors, 5 nurses, with about three weeks of dose history each, so adherence trends look realistic rather than flat).

- **Real reference data for drug interactions** — rather than inventing medicine conflicts, we sourced a set of well-documented, clinically established drug interaction pairs from standard pharmacology references (e.g., American Academy of Family Physicians clinical reviews) — for example, Warfarin with Aspirin (bleeding risk), or Simvastatin with Erythromycin (muscle toxicity risk). This is the one part of the dataset where we deliberately used real, citable medical knowledge instead of fake data, so the interaction-check feature reflects something medically genuine even though the patients themselves are fictional.

Sample Drug Interaction Reference Data

Medicine A	Medicine B	Risk	Severity
Warfarin	Aspirin / Ibuprofen	Increased bleeding risk	Severe
Lisinopril	Spironolactone	Hyperkalemia	Severe
Simvastatin	Erythromycin	Muscle toxicity	Severe
Digoxin	Furosemide	Electrolyte imbalance	Moderate
Sildenafil	Nitroglycerin	Severe hypotension	Severe

Source: standard clinical pharmacology references (e.g., AAFP, "Clinically Relevant Drug-Drug Interactions in Primary Care"). Full list of 10 pairs is seeded in the project's drug_interactions table.

SDG Alignment

SDG 3 — Good Health and Well-being (direct support for medication adherence and preventive care) and **SDG 10 — Reduced Inequalities** (designed with elderly and low-literacy patients in mind, who are disproportionately affected by missed doses and complex schedules).